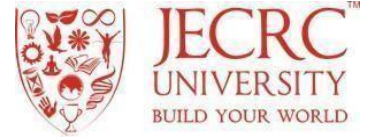


B.Tech.(Computer Science and Engineering)
Semester: III
Subject: DSA(BCO002B)
Marks: 64 Marks



Assignment# 2

Tutorial Sheet 4

SECTION A

(5*2=10)

1. Define stack. What are the applications of stack?
2. What are the base condition and recursive condition in recursion?
3. Convert the infix expression $(A/B * C + D)$ to postfix.
4. State the difference between iteration and recursion.
5. Write the recursive formula for Fibonacci numbers.

SECTION B

(3*7=21)

1. What are the advantages and disadvantages and applications of recursion?.
2. Evaluate the following Postfix expressions using stack ADT:
 - a. $5\ 9\ 3\ +\ 4\ 2\ *\ *\ 7\ +\ *$
 - b. $a\ b\ c\ *\ +\ d\ e\ *\ f\ +\ g\ *\ +$ (where $a=1, b=2, c=3, d=4, e=5, f=6, g=2$)
3. Explain reversing of string using Stack.

SECTION C

(3*11=33)

1. What is recursion? Explain the generation of the Fibonacci series using a recursive function with a suitable example.
2. What is the Tower of Hanoi problem? Explain its recursive solution and show the steps for solving it with 4 disks.
3. Give the algorithm to convert an infix expression into its corresponding prefix expressions. Show the conversion steps for converting the given expression into Prefix using stack ADT.

$$A + (B / C - (D * E ^ F) + G) * H$$